

MODULE 1: Overview of Internet & Application Layer

PCCST501 Computer Networks | Quick Exam Revision Guide | KTU S5 CSE | Prof. Anju Markose, VJCET

1. Overview of the Internet & Protocol Layering

Internet Components: The network is divided into the *Network Edge* (end systems, hosts, clients, servers) and the *Network Core* (mesh of interconnected routers and packet switches).

Packet Switching vs Circuit Switching:

- **Packet Switching:** Discrete data packets share network resources via *statistical multiplexing*. Highly efficient, handles bursty traffic, but introduces queuing delays and packet loss.
- **Circuit Switching:** Dedicated end-to-end communication channels reserved for the entire session (e.g. traditional PSTN phone network). Guaranteed bandwidth, but underutilized when idle.

Key Concept / Exam Formula:

Nodal Delay Formula: Total Delay = $d_{proc} + d_{queue} + d_{trans} + d_{prop}$

- **Transmission Delay (d_{trans}):** L / R (Packet length in bits / Link bandwidth in bps)
- **Propagation Delay (d_{prop}):** d / s (Physical distance in meters / Speed of light $\sim 2 \times 10^8$ m/s)

OSI vs TCP/IP Protocol Layering:

OSI Model (7 Layers)	TCP/IP Model (5 Layers)	Protocol Data Unit (PDU) & Examples
7. Application 6. Presentation 5. Session	5. Application Layer	Message: HTTP, FTP, SMTP, DNS
4. Transport Layer	4. Transport Layer	Segment / Datagram: TCP, UDP
3. Network Layer	3. Network Layer	Packet / Datagram: IPv4, IPv6, ICMP
2. Data Link Layer	2. Data Link Layer	Frame: Ethernet (802.3), WiFi (802.11)
1. Physical Layer	1. Physical Layer	Bits: UTP Cable, Optical Fiber, Radio

2. World Wide Web (WWW) and HTTP

HTTP Characteristics: Stateless client-server protocol over TCP (Default Port 80, HTTPS Port 443).

- **Non-Persistent HTTP (1.0):** Each object download requires a separate TCP connection (2 RTT per object + transmission time).
- **Persistent HTTP (1.1):** Reuses a single TCP connection for multiple web objects, drastically cutting connection establishment latency.
- **HTTP Methods:** GET (retrieve data), POST (submit form/payload), HEAD (retrieve headers only), PUT (upload/replace), DELETE.
- **HTTP Status Codes:** 200 (OK), 301 (Moved Permanently), 304 (Not Modified / Cached), 400 (Bad Request), 404 (Not Found), 500 (Internal Server Error).

- **Cookies & Web Caching:** Cookies provide state management over stateless HTTP via 4 components: Header in HTTP response, Header in next request, Cookie file on client host, Backend database.

3. File Transfer Protocol (FTP) & Electronic Mail

FTP Architecture: Uses two separate parallel TCP connections (Out-of-band control):

- **Control Connection (Port 21):** Carries user commands and server replies throughout the session. Remains open.
- **Data Connection (Port 20):** Opened dynamically for each file transfer and closed immediately after transfer finishes.

Electronic Mail Architecture:

- **SMTP (Simple Mail Transfer Protocol - Port 25/587):** Push protocol between client/server and server/server using direct TCP and 7-bit ASCII commands (HELO, MAIL FROM, RCPT TO, DATA, QUIT).
- **Mail Access Protocols (Pull):** POP3 (Post Office Protocol v3 - Port 110, downloads emails locally, stateless), IMAP4 (Internet Message Access Protocol v4 - Port 143, stateful, synchronizes folders across multiple devices).
- **MIME (Multipurpose Internet Mail Extensions):** Extends SMTP to carry multimedia non-ASCII content using Base64 encoding.

4. Domain Name System (DNS) & P2P / BitTorrent

DNS Architecture: Distributed hierarchical database running over UDP/TCP Port 53.

- **Server Hierarchy:** 13 Root DNS server clusters → Top-Level Domain (TLD) servers (.com, .org, .in) → Authoritative DNS servers.
- **Resolution Types:** *Recursive* (local resolver queries on behalf of client until resolved) vs *Iterative* (resolver receives referral addresses).
- **Resource Records (RR):** Type A (Host → IPv4), Type AAAA (Host → IPv6), Type NS (Domain → Authoritative Name Server), Type CNAME (Canonical Alias), Type MX (Domain → Mail Server).

Peer-to-Peer Paradigm & BitTorrent:

- **P2P Self-Scalability:** As users join, they bring both demand and upload capacity. BitTorrent divides files into 256 KB chunks.
- **Rarest First Policy:** Peers request chunks that are least copied among neighbors first to maximize piece variety.
- **Tit-for-Tat Choking Algorithm:** Peers unchoke top 4 uploaders offering highest rates. Optimistic unchoking tests random new peers every 30 seconds.